

Q1

稳态误差 (ess) = ?

∴ 单位反馈

$H(s) = 1$

∴ $e_{ss} = E_{ss}$

$$\frac{E_N(s)}{N(s)} = \frac{G}{1 + GH}$$

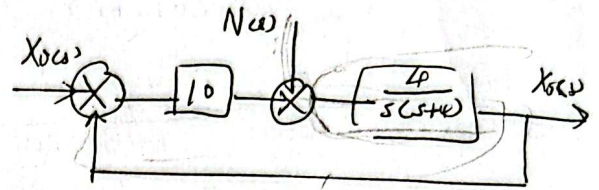
$n(s) = -2.1(s)$

$$|H| = 1, |G| = \frac{4}{s(s+4)}$$

$N(s) = \frac{-2}{s}$

$$\therefore E_N(s) = \frac{\left(\frac{4}{s(s+4)} \cdot 1\right)}{1 + \left(\frac{4}{s(s+4)} \cdot 1\right) \cdot 10} \cdot \left(-\frac{2}{s}\right)$$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} s E_N(s) = \lim_{s \rightarrow 0} \frac{-4}{s^2 + 4s + 40} (-2) = +0.2$$



$$\frac{G}{1 + GH}$$

$$G = \frac{4}{s(s+4)}$$

$$\frac{E_{X_0}(s)}{X_0(s)} = \frac{1}{1 + GH}$$

$$G = 10 \cdot \frac{4}{s(s+4)}$$

$X_0(s) = \frac{1}{s}$

$X_1(s) = \frac{1}{s}$

~~$n(s) = -2.1(s)$~~

~~$N(s) = \frac{-2}{s}$~~

$$\therefore E_{X_0}(s) = \frac{1}{1 + 10 \cdot \frac{4}{s(s+4)}} \left(\frac{1}{s}\right)$$

$$e_{ss} = \lim_{s \rightarrow 0} s E_{X_0}(s) = \lim_{s \rightarrow 0} \frac{s(s+4)}{s^2 + 4s + 40} \left(\frac{1}{s}\right) = +0.1$$

$$\therefore e_{ss} = e_{ss_N} + e_{ss_{X_1}} = 0.3$$

开环 单位反馈

Q2

$$G(s) = \frac{100}{s(0.1s+1)}$$

$$x_o(t) = (1 + t + at^2) \cdot 1(t), \quad a \geq 0$$

解 a

$$\begin{aligned} \frac{E(s)}{X_o(s)} &= \frac{1}{1+G(s)H(s)}, \\ &= \frac{1}{1+G(s)} \\ &= \frac{s(0.1s+1)}{0.1s^2 + s + 100} \end{aligned}$$

$$H(s) = 1, \quad \therefore e_{ss} = E_{ss}$$

$$x_o(t) = (1 + t + at^2) \cdot 1(t)$$

$$X_i(s) = \frac{1}{s} + \frac{1}{s^2} + \frac{2a}{s^3} = \frac{s^2 + s + 2a}{s^3}$$

$$E(s) = \left(\frac{E(s)}{X_o(s)} \right) \cdot X_o(s)$$

$$e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \frac{E(s)}{X_o(s)} \cdot X_o(s)$$

$$\therefore e_{ss} = \infty \quad (a > 0)$$

$$e_{ss} = 0.1 \quad (a = 0)$$

Q3 c1) $X_o(s) = 1(s)$, $ess = ?$

$$G_a = \frac{G}{1+GH} \quad , \quad H = K_h$$

$$= \frac{\frac{K}{Js+F}}{1 + \frac{K \cdot K_h}{Js+F}}$$

$$= \frac{K}{(Js+F) + K \cdot K_h}$$

$$\frac{E(s)}{X_o(s)} = \frac{1}{1 + (\frac{1}{s} G_a) H'} \quad , \quad H' = 1$$

$$= \frac{(\frac{1}{s} G_a)}{\frac{1}{s} [(Js+F) + K \cdot K_h]}$$

$$= \frac{s(Js+F+K \cdot K_h)}{Js^2 + (F+K \cdot K_h)s + K}$$

$\therefore X_o(s) = 1(s)$, $\therefore X_o(s) = \frac{1}{s}$

$$ess(\infty) = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \left(\frac{E(s)}{X_o(s)} \right) \cdot X_o(s) = 0$$

(2) $X_o(s) = \frac{1}{s^2}$ (单位斜坡)

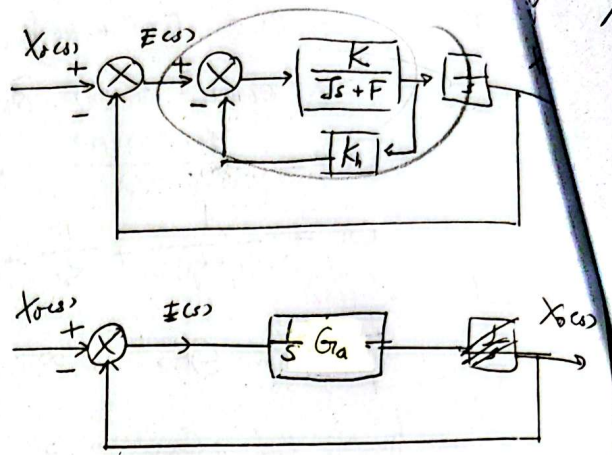
$$X_o(s) = \frac{1}{s^2}$$

$$ess(\infty) = \lim_{s \rightarrow 0} s \frac{E(s)}{X_o(s)} X_o(s) = \frac{F + K \cdot K_h}{K} = \frac{F}{K} + K_h$$

c3) K_h 对 K , ess 影响, $\therefore ess = ess_1 + ess_2$

K_h 对 ess_1 无影响,

$K_h \nearrow \quad ess_1 \nearrow$, $K_h \searrow \quad ess_2 \searrow$ ($\therefore ess$ 与 K_h 正相关)



阶跃 斜坡 加速
 $\theta_0(t) = [10 + 60t(1+t)] \cdot 1(t)$

$e(t) = \theta_0(t) - \theta_o(t)$

$G(s) = \frac{400(0.3s+1)}{s^2(s+1)} \quad k=2 \text{ (II-)}$

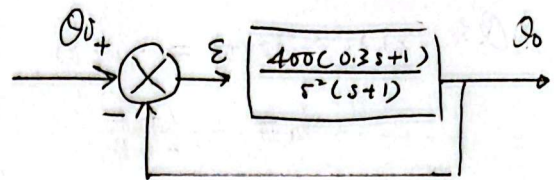
$e_{ss} = \frac{A}{1+k_p} + \frac{B}{k_v} + \frac{C}{k_a} \quad \begin{cases} A = 10 \\ B = 60 \\ C = 60 \end{cases}$

$k_p = \lim_{s \rightarrow 0} G(s) \rightarrow \infty$

$k_v = \lim_{s \rightarrow 0} sG(s) \rightarrow \infty$

$k_a = \lim_{s \rightarrow 0} s^2 G(s) = \lim_{s \rightarrow 0} \frac{400(0.3s+1)}{s+1} = 400$

$\therefore e_{ss} = 0 + 0 + \frac{60}{400} = 0.15$



Q5 $x_0(t) = \theta_m \sin(\omega t)$, $\omega_{max} = 0.5 \text{ rad/s}$
 $\varepsilon_{max} = 1 \text{ rad/s}^2$, $\nabla |E|_{max}$

$\dot{x}_0(t) = \dot{\theta}_0(t) = \theta_m \omega \cos \omega t$

$\ddot{x}_0(t) = \ddot{\theta}_0(t) = -\theta_m \omega^2 \sin \omega t$

$\begin{cases} \theta_m \omega = 0.5 \\ \theta_m \omega^2 = 1 \end{cases}$

$\therefore \begin{cases} \omega = 2 \text{ rad/s} \\ \theta_m = 0.25 \text{ rad} \end{cases}$

$\theta_0(t) = x_0(t) = 0.25 \sin(2t)$

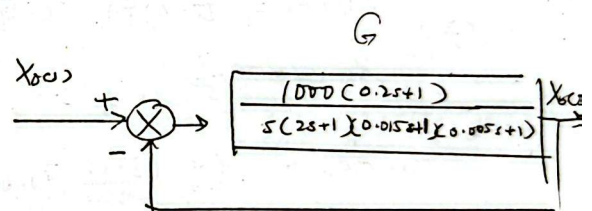
$\frac{1}{1+GH} = \frac{E(s)}{\theta_0(s)} = \frac{1}{1 + \frac{1000(0.2s+1)}{s(2s+1)(0.005s+1)}}$
 $\left| \frac{E(j\omega)}{\theta_0(j\omega)} \right| = \frac{\omega \sqrt{(2\omega)^2+1} \sqrt{(0.005\omega)^2+1}}{\sqrt{1 + \frac{1000(0.2\omega+1)}{s(2s+1)(0.005s+1)}}}$

$\theta_0(t)_{max} = 0.25$ $\omega(t) = \theta_m \omega \cos \omega t$

$\ddot{\theta}_0(t)_{max} = \theta_m \omega^2$

$\omega(t) = 2 \text{ rad/s}$

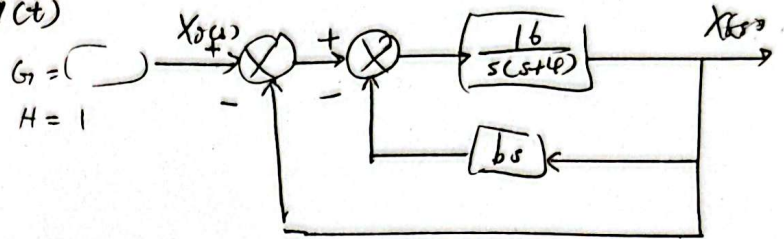
$\therefore |E(j\omega)| = 2.08 \times 10^{-3} \text{ rad}$



c1) $b=0$, $x_1(t) = t \cdot 1(t)$

$$\frac{F(s)}{X_0(s)} = \frac{1}{1 + G_1 H}$$

$$= \frac{s(s+4)}{s(s+4) + 16}$$



$\therefore x_0(t) = t \cdot 1(t)$, $X_{0(s)} = \frac{1}{s^2}$

~~$\lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \left(\frac{s^2(s+4)}{s(s+4) + 16} \right)$~~

$\lim_{s \rightarrow 0} s \frac{F(s)}{X_{0(s)}} \cdot X_{0(s)} = \lim_{s \rightarrow 0} \frac{s(s+4)}{s[s(s+4) + 16]} = \frac{0+4}{0+16} = 0.25$

c2) $b = 0.15$,

$G_a = \frac{G}{1 + G_1 H}$, $H = bs = 0.15s$

$\therefore = \frac{\frac{1}{s(s+4)}}{1 + \frac{16}{s(s+4)}(0.15s)}$

$\therefore \frac{F(s)}{X_{0(s)}} = \frac{s(s+6.4)}{s(s+6.4) + 16}$

$\therefore X_{0(s)} = \frac{1}{s^2}$

~~$\lim_{s \rightarrow 0} s E(s)$~~ $\lim_{s \rightarrow 0} s \frac{F(s)}{X_{0(s)}} \cdot X_{0(s)} = \lim_{s \rightarrow 0} \frac{s+6.4}{s[s(s+6.4) + 16]} = 0.4$

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Q4-4 (11-2) (已知) $\alpha = 20^\circ, h_a^* = 1, m = 4\text{mm},$
 $z_1 = 18, z_2 = 41,$

(分度圆) $\therefore d = mz, r = \frac{mz}{2}, \therefore \begin{cases} r_1 = \frac{4(18)}{2} = 36\text{mm} \\ r_2 = \frac{4(41)}{2} = 82\text{mm} \end{cases}$

(base: 基圆) $d = m\cos\alpha, r_b = \frac{mz\cos\alpha}{2}, \therefore \begin{cases} r_{b1} = 33.829\text{mm} \\ r_{b2} = 77.055\text{mm} \end{cases}$

(foot: 齿顶圆) $d_f = m(z - 2h_a^* - 2c^*)$
 $r = \frac{m}{2}(z - 2h_a^* - c^*) \therefore \begin{cases} r_{f1} = 31\text{mm} \\ r_{f2} = 77\text{mm} \end{cases}$

$\therefore \begin{cases} h_a^* = 1, c^* = 0.25 \\ m \geq 1\text{mm} \end{cases}$

(齿顶) $d_a = m(z + 2h_a^*), r = \frac{m}{2}(z + 2h_a^*) \therefore \begin{cases} r_{a1} = 40\text{mm} \\ r_{a2} = 86\text{mm} \end{cases}$ (平行轴斜齿轮)

$\varepsilon_\gamma = \varepsilon_\alpha + \varepsilon_\beta$

$a = \frac{m}{2}(z_1 + z_2) = r_1 + r_2 = 118\text{mm}$
 ε_α

$\varepsilon_\gamma = \frac{1}{2\pi} [z_1 (\tan \alpha_{a1} - \tan \alpha') + z_2 (\tan \alpha_{a2} - \tan \alpha')] + \frac{b \sin \beta}{\pi m_n}$

$\cos \alpha_{a1} = \frac{r_{b1}}{r_{a1}}, \therefore \alpha_{a1} = 32.25^\circ, \cos \alpha_{a2} = \frac{r_{b2}}{r_{a2}}, \alpha_{a2} = 26.36^\circ$

$\therefore \tan \alpha_{a1} = 0.631$

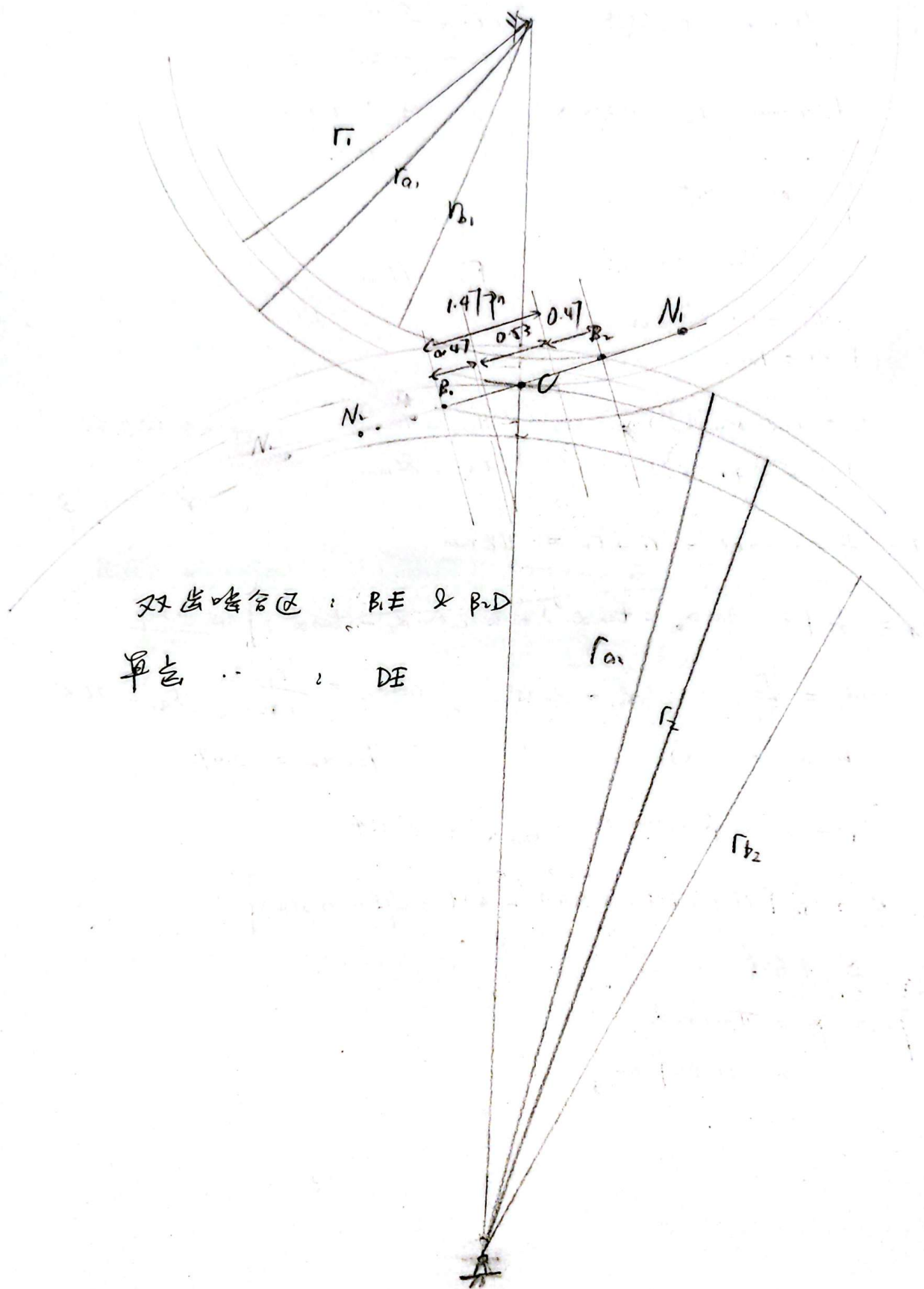
$\therefore \tan \alpha_{a2} = 0.496$

$\alpha' = 20^\circ$ (分度圆压力角), $\tan \alpha' = 0.364$

$\therefore \varepsilon_\alpha = \frac{1}{2\pi} [18(0.631 - 0.364) + 41(0.496 - 0.364)]$
 ≈ 1.626

$p_n = p_b = \pi m \cos \alpha$
 $= 11.809\text{mm}$

Q 4.4 (2) ($n=2$) 面理论, 例啮合 $N_1 N_2$, 实际 — $B_1 B_2$



双齿啮合区: $B_1 E$ & $B_2 D$

单齿 ... DE

$$z_1 = 17, z_2 = 118, m = 5 \text{ mm}, h_a^* = 1, a' = 337.5 \text{ mm}$$

$$\delta = 0.91 \text{ mm} \quad (\text{按 } \delta < 0.91)$$

$$\therefore \delta = \frac{\pi m}{2} + 2xm \tan \alpha$$

$$\delta = 2xm \tan \alpha, \quad \therefore m = 5, \alpha = 20^\circ$$

$$\frac{0.91}{2(5) \tan 20^\circ} = x_1, \quad \therefore x_1 = -0.25$$

$$\therefore x_1 + x_2 = 0 \quad (\text{零传动}), \quad \therefore x_2 = 0.25$$

$$\therefore r_1 = \frac{m z_1}{2} = 42.5 \text{ mm}$$

$$r_2 = \frac{m z_2}{2} = 295 \text{ mm}$$

$$r_{b1} = r_1 \cos \alpha = 39.937 \text{ mm}$$

$$r_{b2} = r_2 \cos \alpha = 277.209 \text{ mm}$$

$$r_{f1} = r_1 - m(h_a^* + c^* - x_1) = 37.5 \text{ mm}$$

$$r_{f2} = r_2 - m(h_a^* + c^* - x_2) = 287.5 \text{ mm}$$

$$r_{a1} = r_1 + m(h_a^* + x_1) = 48.75 \text{ mm}$$

$$r_{a2} = r_2 + m(h_a^* + x_2) = 298.75 \text{ mm}$$

$$E_a = \frac{1}{2\pi} \left[z_1 (\tan \alpha_{a1} - \tan \alpha) + z_2 (\tan \alpha_{a2} - \tan \alpha) \right]$$

$$= 1.62$$

$$(\text{正变位齿顶厚}) \quad S_a = s \left(\frac{r_a}{r} \right) - 2r_a (\text{inv } \alpha_a - \text{inv } \alpha), \quad \text{其中 } \text{inv } \alpha = \tan \alpha - \alpha \quad (\text{BSW})$$

$$S_{a1} = 2.795 \text{ mm}$$

$$S_{a2} = 4.1418 \text{ mm}$$

$$\therefore d = mz$$

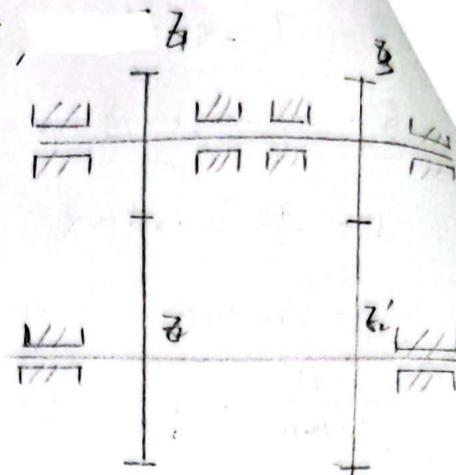
$$d_b = mz \cos \alpha$$

$$d_a = (z + 2h_a^*)m$$

$$\therefore (z, h_a^*, c^*)$$

Q4-6 $z_1 = 27, z_2 = 60, z'_2 = 63, z_3 = 25,$
 $m = 4 \text{ mm}, \alpha = 20^\circ$

(传动设计方第)



~~$a = 174 \text{ mm}$~~

$$a' = r_1 + r_2 = \frac{m z_1}{2} + \frac{m z_2}{2} = 174 \text{ mm}$$

$$a'' = r_3 + r'_2 = \frac{m z_3}{2} + \frac{m z'_2}{2} = 176 \text{ mm}$$

方案①: ~~$a = 174 \text{ mm}$~~ ~~$a = 176$~~ 两齿轮皆为负传动 $\begin{cases} x_1 + x_2 < 0 \\ x_3 + x'_2 < 0 \end{cases}$
 $a < 174 < 176$

方案②: $a = 174 < 176$, z_1, z_2 标准变齿, z_3, z'_2 负传动 $\begin{cases} x_1 + x_2 = 0 \\ x_3 + x'_2 < 0 \end{cases}$

方案③: $174 < a < 176$, z_1, z_2 正传动, z_3, z'_2 负传动 $\begin{cases} x_1 + x_2 > 0 \\ x_3 + x'_2 < 0 \end{cases}$

方案④: $174 < a = 176$ ~~$a = 174$~~ z_1, z_2 正, z_3, z'_2 标准 $\begin{cases} x_1 + x_2 > 0 \\ x_3 + x'_2 = 0 \end{cases}$

方案⑤: $a > 176 > 174$ 两齿轮皆为正传动 $\begin{cases} x_1 + x_2 > 0 \\ x_3 + x'_2 > 0 \end{cases}$

综上, 选择方案⑤, 正传动可提升齿轮抗弯能力, 减少磨损。

Q4-7 $m = 4 \text{ mm}$, $\alpha = 20^\circ$, $h_a^* = 1$, $z_1 = 23$, $z_2 = 47$
 $z_1' = 23$, $z_2' = 46$ (不重合)

$$r_1' = \frac{m z_1'}{2}, \quad r_2' = \frac{m z_2'}{2}$$

$$= 46 \text{ mm} \quad = 92 \text{ mm}$$

~~$s = \frac{\pi m}{2} + 2x$~~

$$a = r_1' + r_2' = 138 \text{ mm}$$

$$a' = r_1 + r_2 = 140 \text{ mm}$$

$$a \cos \alpha = a' \cos \alpha'$$

$$\therefore \alpha' = \arccos\left(\left(\frac{a}{a'}\right) \cos \alpha\right)$$

$$\approx 22.14^\circ$$

$$\tan \alpha' = \frac{2(z_1 + z_2)}{z_1 + z_2'} \tan \alpha + \frac{m}{a'} \alpha$$

不重合, $x_2 = 0$

$$x_1 + x_2 = \frac{z_1 + z_2'}{2 + \tan \alpha} (\tan \alpha' - \tan \alpha)$$

$$\therefore x_1 = 0.5287$$

$$y = \frac{a' - a}{m} = \frac{140 - 138}{4} = \frac{1}{2}$$

$$\Delta y = x_1 + x_2 - y = 0.0287$$

$$h_{a1} = (h_a^* + x_1 - \Delta y)m = 6 \text{ mm}$$

$$r_{a1} = r_1 + h_{a1} = 52 \text{ mm}$$

$$h_{a2} = (h_a^* + x_2 - \Delta y)m = 3.852 \text{ mm}$$

$$r_{a2} = r_2 + h_{a2} = 95.852 \text{ mm}$$

$$h_{f1} = (h_a^* + x_1 - \Delta y)m = 2.852 \text{ mm}$$

$$\alpha_{a1} = \cos^{-1}\left(\frac{r_{b1}}{r_{a1}}\right) = 33.77^\circ$$

$$h_{f2} = (h_a^* + x_2 - \Delta y)m = 5 \text{ mm}$$

$$\alpha_{a2} = \cos^{-1}\left(\frac{r_{b2}}{r_{a2}}\right) = 25.63^\circ$$

$$\varepsilon_a = \frac{1}{\pi} \left[z_1 (\tan \alpha_{a1} - \tan \alpha') + z_2 (\tan \alpha_{a2} - \tan \alpha') \right]$$

$$= 1.492$$

$$S_{a1} = S_1 \left(\frac{r_{a1}}{r_f} \right) - 2r_{a1} (\tan \alpha_{a1} - \tan \alpha')$$

$$= 2.147 \text{ mm}$$

$$p_n = \pi m \cos \alpha = 11.8085 \text{ mm}$$

~~11.8085~~

